

Coordinate Systems

Cartesian Coordinates

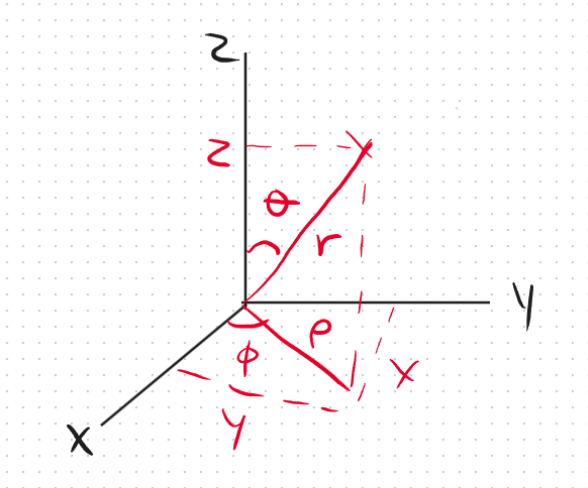
Based on the $\hat{x}$, $\hat{y}$, and $\hat{z}$ unit vectors. Format is (x, y, z)

Cylindrical Coordinates

Based on the $\hat{\rho}$, $\hat{\phi}$, and $\hat{z}$ unit vectors. Format is (ρ, ϕ, z)

Spherical Coordinates

Based on the $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$ unit vectors. Format is (r, θ, ϕ)



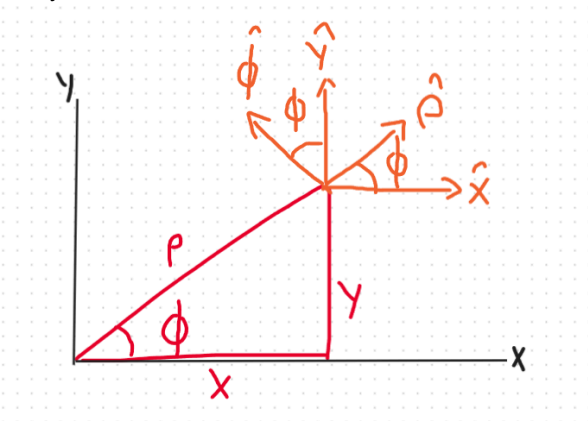
Coordinate System Equations

$$\begin{aligned} x &= \rho \cos \phi & y &= \rho \sin \phi & z &= r \cos \theta \\ r &= \sqrt{x^2 + y^2 + z^2} & \rho &= \sqrt{x^2 + y^2} & \rho &= r \sin \theta \end{aligned}$$

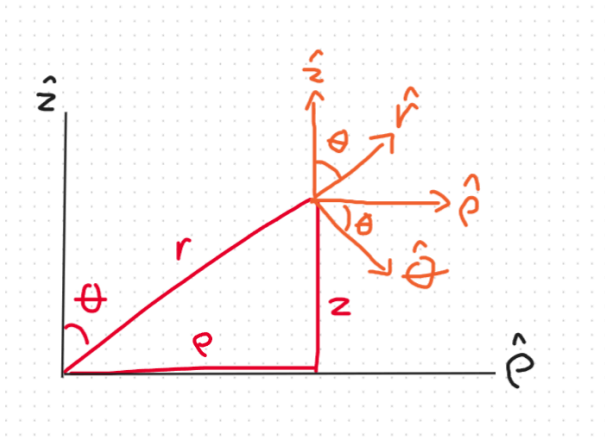
Note that ϕ always starts from $+x$, CCW; θ always starts from $+z$, going down to ρ .

Unit Vectors

For cylindrical coordinates:



$$\begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{\phi} \end{bmatrix} \quad \begin{bmatrix} \hat{\rho} \\ \hat{\phi} \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$



$$\begin{bmatrix} \hat{\theta} \\ \hat{r} \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{z} \end{bmatrix} \quad \begin{bmatrix} \hat{\rho} \\ \hat{z} \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \hat{\theta} \\ \hat{r} \end{bmatrix}$$

This implies that the non-cartesian coordinates are position dependent, and so their unit vectors are dependent on the position of the point.

Gradient of Cylindrical and Spherical

Given generalized coordinates (q_1, q_2, q_3) with position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = \vec{r}(q_1, q_2, q_3)$:

$$\vec{\nabla} f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3}$$

where $h_1 = \left| \frac{\partial \vec{r}}{\partial q_1} \right|$ and the same pattern for the rest of the partial derivatives.

Table of Coordinates

Generalized (Curvilinear)	h_1	q_1	$\hat{e}_1$	h_2	q_2	$\hat{e}_2$	h_3	q_3	$\hat{e}_3$
Cartesian	1	x	$\hat{x}$	1	y	$\hat{y}$	1	z	$\hat{z}$
Cylindrical	1	ρ	$\hat{\rho}$	ρ	ϕ	$\hat{\phi}$	1	z	$\hat{z}$
Spherical	1	r	$\hat{r}$	r	θ	$\hat{\theta}$	$r \sin \theta$	ϕ	$\hat{\phi}$

Divergence and Curl of a Generalized Coordinate Vector

Given generalized coordinates (q_1, q_2, q_3) with position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = \vec{r}(q_1, q_2, q_3)$:

$$\vec{\nabla} \cdot \vec{v} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 v_1) + \frac{\partial}{\partial q_2} (h_3 h_1 v_2) + \frac{\partial}{\partial q_3} (h_1 h_2 v_3) \right]$$

and

$$\vec{\nabla} \times \vec{v} = \begin{vmatrix} h_1 \hat{e}_1 & h_2 \hat{e}_2 & h_3 \hat{e}_3 \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix}$$